COMPLETE MATHEMATICS HANDBOOK • REVISION 2021

Mathematics II

A complete student textbook for determinants, matrices, vector algebra, integral calculus, applications of integration and first-order differential equations—with methods, diagrams, solved problems, practice, quick revision and full model answers.

Course Code
2002

Programme
Diploma in Engineering and Technology

Semester
2

Credits
4

Category
Basic Science

Periods
4/week • 60/semester

Student promise
Every official syllabus item is developed through concepts, methods, worked examples and examination practice. The handbook does not add advanced topics beyond the level needed to understand and apply the prescribed syllabus.

$$\begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$$

Four-module learning path

Determinants, matrices, Cramer’s rule and inverse method

Vectors, dot product, cross product, work and moment

Indefinite and definite integration

Area and first-order differential equations

COURSE OVERVIEW

Mathematics as an engineering problem-solving language

Mathematics II continues the algebra and calculus foundation developed in Mathematics I. It teaches four connected tools: matrices for simultaneous systems, vectors for magnitude and direction, integration for accumulation, and differential equations for changing systems.

Linear systems

Determinants and matrices organise several equations and provide systematic methods such as Cramer’s rule and inverse matrices.

Magnitude and direction

Vectors describe force, displacement, velocity and moment. Dot and cross products translate geometry into calculation.

Accumulation and change

Integration finds accumulated quantity and area. Differential equations describe relations involving a quantity and its rate of change.

Where diploma students use these ideas

Electrical networks

Simultaneous equations for currents and voltages.

Mechanics

Force resolution, work and turning moment.

Geometry and design

Area enclosed by curves and coordinate axes.

Course 2002

Dynamic systems

Simple growth, decay and linear first-order models.

HOW TO USE THIS HANDBOOK

Understand the method before memorising the formula

1. Read

Begin with the meaning, notation and conditions of the concept.

2. Follow

Trace each worked example line by line, especially signs and limits.

3. Practise

Close the solution and repeat the problem independently.

4. Check

Use substitution, differentiation, dimensions or geometric sense to verify.

Examination method

Write the formula or method first, show essential steps, simplify carefully, box the final answer, and include units for work, moment and engineering quantities.

TABLE OF CONTENTS

Clickable handbook map

OFFICIAL SYLLABUS ANALYSIS

Course objective and prerequisite

Official objective

To give a comprehensive coverage at an introductory level to Determinants and Matrices, Integral Calculus, basic elements of Vector Algebra and First Order Differential Equations as applicable to solve engineering problems.

Student-friendly meaning

The course is not about memorising isolated formulas. It develops four practical methods that help organise equations, represent directed quantities, calculate accumulation and model change.

Prerequisite

Course: Mathematics I, Semester 1

Required topics: Complex Numbers, Coordinate Geometry, Trigonometry, Limits and Derivatives.

Derivatives
needed to understand integration as the reverse operation.

Coordinate geometry
needed for vectors, position vectors and areas.

Trigonometry
needed for vector angles and standard integrals.

Algebra
needed for determinants, matrices and differential equations.

OFFICIAL COURSE OUTCOMES

What students must be able to do

CO	Official outcome	Duration	Cognitive level	Student meaning
CO1	Make use of Determinants and Matrices in finding the solutions of a linear system.	12 h	Applying	Evaluate determinants and solve simultaneous equations.
CO2	Identify the concept of scalar and vector quantities and apply it in engineering problems.	16 h	Applying	Use vector operations for work, angle and moment.
CO3	Build the concept of integration as the inverse operation of differentiation.	18 h	Applying	Integrate using standard results, substitution, parts and limits.
CO4	Apply integration techniques to solve different engineering problems and differential equations.	12 h	Applying	Find area and solve simple first-order differential equations.
—	Series tests	2 h	—	One hour after Module 2 and one hour after Module 4.

CO-PO mapping

Course outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	Interpretation
CO1	3	—	—	—	—	—	—	Strongly mapped to PO1
CO2	3	—	—	—	—	—	—	Strongly mapped to PO1
CO3	3	—	—	—	—	—	—	Strongly mapped to PO1
CO4	3	—	—	—	—	—	—	Strongly mapped to PO1

Mapping scale
3 - Strongly mapped; 2 - Moderately mapped; 1 - Weakly mapped.

OFFICIAL MODULE ROADMAP

Module outcomes, hours and level

Module outcome	Description	Hours	Level
M1.01	Outline the second and third order determinants.	5	Understanding
M1.02	Apply Cramer’s Rule to solve a linear system of equations.	2	Applying
M1.03	Solve a system of equations in two unknowns by inverse matrix.	5	Applying
M2.01	Compare vector and scalar quantities.	4	Understanding
M2.02	Explain vector operations, scalar product and vector product.	6	Understanding
M2.03	Apply dot and cross products to work and moment.	6	Applying
Series Test I	Modules 1 and 2	1	—
M3.01	Relate integration to reverse differentiation and rules of integration.	3	Understanding
M3.02	Solve integration problems by substitution and by parts.	8	Applying
M3.03	Explain definite integrals and solve problems.	7	Applying
M4.01	Apply integration to find area bounded by a curve and the axes.	7	Applying
M4.02	Outline and solve first-order, first-degree differential equations.	5	Applying
Series Test II	Modules 3 and 4	1	—

MODULE 1 • CO1 • 12 HOURS

Determinants and Matrices

Second- and third-order determinants, matrix operations, Cramer’s rule for three variables, inverse of a matrix, and inverse method for two variables.

1. Determinants: meaning, order and notation

A determinant is a single number associated with a square arrangement of numbers. It helps test whether a linear system has a unique solution and is used in inverse matrices and Cramer’s rule.

$$\text{Second order: } |A| = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

Malayalam Support: Determinant ഒരു square array-യിൽ നിന്ന് ലഭിക്കുന്ന ഒറ്റ സംഖ്യയാണ്. Equations-ന് unique solution ഉണ്ടോ എന്ന് പരിശോധിക്കാനും inverse കണ്ടെത്താനും ഇത് ഉപയോഗിക്കുന്നു.

Order

A determinant with n rows and n columns has order n. A 2×2 determinant is second order; a 3×3 determinant is third order.

Rows and columns

Horizontal entries form rows; vertical entries form columns. Determinants are defined only for square arrays.

Solved Example 1: Second-order determinant

Evaluate | 4 3; 2 5 |.

$$|A| = (4\times5) - (3\times2) = 20 - 6 = 14$$

Answer: 14.

2. Third-order determinants

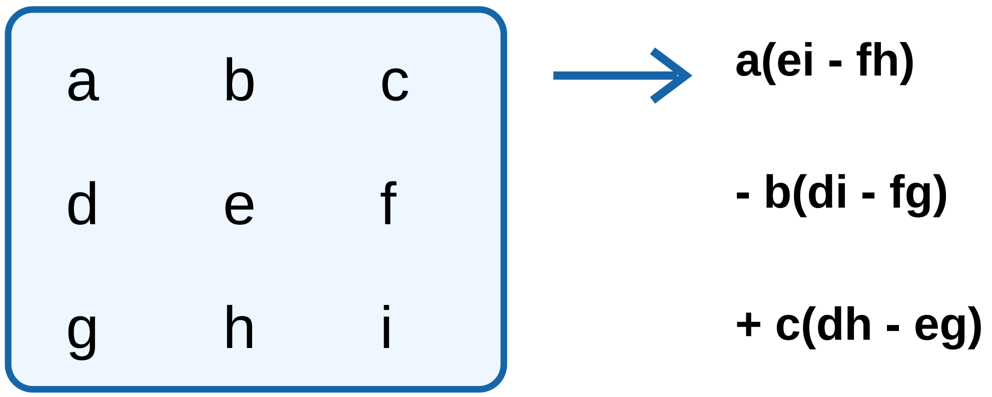
A third-order determinant may be expanded along any row or column. Signs of cofactors follow the checkerboard pattern:

$$\text{Sign pattern} = \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

$$\text{For } A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}, |A| = a(ei-fh) - b(di-fg) + c(dh-eg)$$

Common mistake

The middle term is negative. Write the sign pattern before expanding.



Expansion along the first row. Each entry multiplies the determinant left after deleting its row and column.

Solved Example 2: Third-order determinant

Evaluate $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 2 & 0 & 1 \end{bmatrix}$.

$$\Delta = 1(1\times1-4\times0) - 2(0\times1-4\times2) + 3(0\times0-1\times2)$$

$$\Delta = 1 - 2(-8) + 3(-2) = 1+16-6 = 11$$

Answer: 11.

3. Matrices

A matrix is a rectangular arrangement of numbers in rows and columns. Unlike a determinant, a matrix itself is an array, not a single numerical value.

$$A = [a_{ij}]_{m\times n} \text{ means m rows and n columns.}$$

Type	Definition	Example
Row matrix	One row	[2 5 -1]
Column matrix	One column	[2; 5; -1]
Rectangular	Rows ≠ columns	2×3 matrix
Square	Rows = columns	2×2 or 3×3
Zero/null	Every entry is zero	O
Diagonal	Non-diagonal entries are zero	diag(2,3)
Scalar	Diagonal entries equal	diag(4,4)
Identity/unit	Diagonal entries 1; others 0	I
Upper/lower triangular	Entries below/above main diagonal are zero	Square triangular matrix

4. Equality, addition and subtraction

Two matrices are equal only when they have the same order and every corresponding entry is equal. Addition and subtraction are possible only for matrices of the same order.

$$[a_{ij}] \pm [b_{ij}] = [a_{ij} \pm b_{ij}]$$

Solved Example 3: Matrix addition

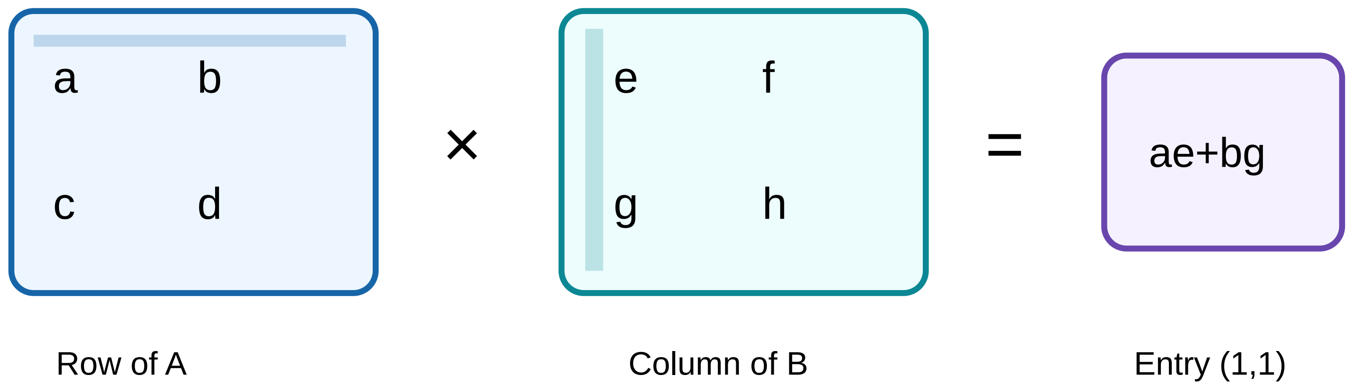
Let A= $\begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}$, B= $\begin{bmatrix} 1 & 5 \\ -2 & 0 \end{bmatrix}$.

$$A+B = \begin{bmatrix} 3 & 4 \\ 1 & 4 \end{bmatrix}; A-B = \begin{bmatrix} 1 & -6 \\ 5 & 4 \end{bmatrix}$$

5. Matrix multiplication

The product AB exists when the number of columns of A equals the number of rows of B. If A is m×n and B is n×p, then AB is m×p.

$(AB)_{ij}$ = sum of products of row i of A with column j of B.



Row-by-column rule. Matrix multiplication is generally not commutative: AB may not equal BA.

Solved Example 4: Matrix product

A= $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, B= $\begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix}$.

$$AB = \begin{bmatrix} 1\times2+2\times1 & 1\times0+2\times5 \\ 3\times2+4\times1 & 3\times0+4\times5 \end{bmatrix} = \begin{bmatrix} 4 & 10 \\ 10 & 20 \end{bmatrix}$$

6. Inverse of a 2×2 matrix

A square matrix A has an inverse only when $|A|\neq0$. Such a matrix is nonsingular.

$$\text{If } A=\begin{bmatrix} a & b \\ c & d \end{bmatrix}, \text{ then } A^{-1} = 1/(ad-bc) \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Solved Example 5: Inverse

Find the inverse of A= $\begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$.

$|A|=3\times1-1\times2=1$.

$$A^{-1}=\begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$$

Check: $AA^{-1}=I$.

7. Solving two equations by inverse matrix

$$AX=B \Rightarrow X=A^{-1}B$$

Solved Example 6: Inverse method

Solve 3x+y=7 and 2x+y=5.

$$A=\begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix},\ X=\begin{bmatrix} x \\ y \end{bmatrix},\ B=\begin{bmatrix} 7 \\ 5 \end{bmatrix}$$

$$X=A^{-1}B=\begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}\begin{bmatrix} 7 \\ 5 \end{bmatrix}=\begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Answer: x=2, y=1.

8. Cramer’s rule for three variables

For a system AX=B with Δ=|A|≠0, replace one coefficient column at a time by the constants column.

$$x=\Delta_x/\Delta,\ y=\Delta_y/\Delta,\ z=\Delta_z/\Delta$$

1. Write coefficients in the same variable order.
2. Evaluate coefficient determinant Δ.
3. Form Δx, Δy and Δz by replacing the corresponding column.
4. Divide each replaced determinant by Δ.
5. Substitute values into the original equations to verify.

Solved Example 7: Cramer’s rule

Solve x+y+z=6, 2x-y+z=3, x+2y-z=2.

Coefficient determinant:

$$\Delta=\begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 1 & 2 & -1 \end{bmatrix}=7$$

$$\Delta x=7,\ \Delta y=14,\ \Delta z=21$$

Answer: x=1, y=2, z=3.

Module 1 exam tip

For Cramer’s rule, preserve the variable order x, y, z in every determinant. For inverse method, first confirm $|A|\neq 0$.

Module 1 Quick Check and Answers

1. Evaluate |5 2; 3 1|. **Answer:** -1.
2. Can a 2×3 matrix have a determinant? **Answer:** No.
3. State the order of AB if A is 2×3 and B is 3×4. **Answer:** 2×4.
4. When does A^{-1} exist? **Answer:** When A is square and $|A|\neq 0$.

MODULE 2 • CO2 • 16 HOURS • SERIES TEST I

Vector Algebra

Scalars and vectors, types of vectors, algebra, position vector, scalar product, vector product, work and moment.

1. Scalars and vectors

Scalar

A quantity having magnitude only. Examples: mass, time, temperature, energy and speed.

Vector

A quantity having magnitude and direction and obeying vector addition. Examples: displacement, velocity, acceleration, force and moment.

Malayalam Support: Scalar-ന് magnitude മാത്രം. Vector-ന് magnitude-ഉം direction-ഉം ഉണ്ട്. Force, displacement എന്നിവ vectors ആണ്.

2. Vector notation, magnitude and unit vector

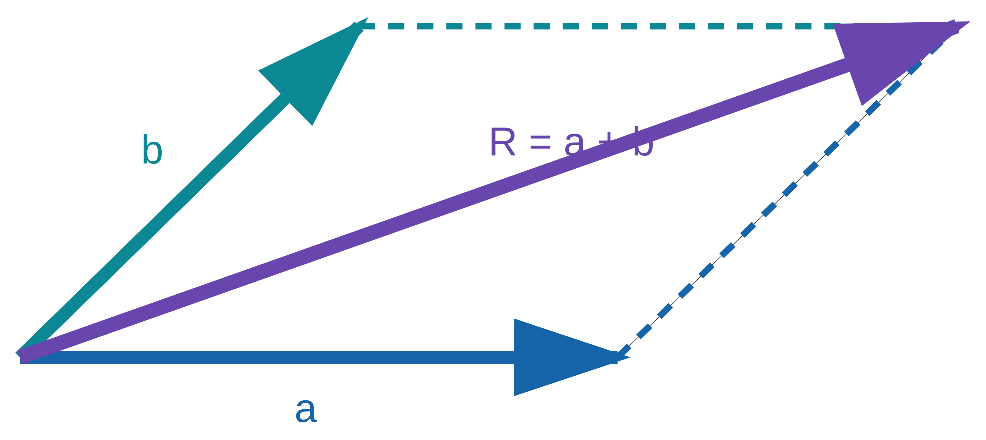
$$a = a_x i + a_y j + a_z k$$

$$|a| = \sqrt{(a_x^2 + a_y^2 + a_z^2)}, \quad \hat{a} = a/|a|$$

Type	Meaning
Zero vector	Magnitude zero; direction not definite.
Unit vector	Magnitude one.
Equal vectors	Same magnitude and same direction.
Negative vectors	Same magnitude, opposite directions.
Parallel/collinear	One is a scalar multiple of the other.
Like/unlike vectors	Parallel vectors with same/opposite direction.
Position vector	Vector from origin to a point.

3. Vector addition and subtraction

$$a+b=(a_x+b_x)i+(a_y+b_y)j+(a_z+b_z)k$$



Parallelogram law: the diagonal from the common tail represents the resultant a+b.

Solved Example 8: Magnitude and unit vector

For a=3i-4j+12k, |a|=√(9+16+144)=13.

$$\hat{a}=(3/13)i-(4/13)j+(12/13)k$$

4. Position vector

$$\text{If } P(x,y,z),\ OP = xi+yj+zk.$$

$$\text{Vector } AB = OB-OA = (x_2-x_1)i+(y_2-y_1)j+(z_2-z_1)k$$

Solved Example 9: Vector between points

A(1,2,-1), B(4,-2,3).

$$AB=(4-1)i+(-2-2)j+(3+1)k=3i-4j+4k$$

$|AB|=\sqrt{(9+16+16)}=\sqrt{41}$.

5. Scalar or dot product

$$a\cdot b = |a||b|\cos\theta = a_x b_x + a_y b_y + a_z b_z$$

Angle

$$\cos\theta=(a\cdot b)/(|a||b|)$$

Perpendicular test

For nonzero vectors, $a\cdot b=0$ means a is perpendicular to b.

Solved Example 10: Dot product and angle

$a=2i+j-2k$, $b=i+2j+2k$.

$a\cdot b=2+2-4=0$.

Conclusion: The vectors are perpendicular.

6. Work done by a constant force

$$\text{Work } W = F\cdot s = |F||s|\cos\theta$$

Only the component of force along the displacement contributes to work.

Solved Example 11: Work

A force $F=4i-2j+3k$ N moves a body through $s=2i+j-4k$ m.

$$W=F\cdot s=4\times 2+(-2)\times 1+3\times(-4)=8-2-12=-6\text{ J}$$

Answer: -6 J. Negative work means the force has a component opposing displacement.

7. Vector or cross product

$$a\times b = |a||b|\sin\theta\ \hat{n}$$

$$a\times b = \begin{bmatrix} i & j & k \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{bmatrix}$$

Direction
The direction is perpendicular to both a and b and follows the right-hand rule. Also a×b=-(b×a).

Solved Example 12: Cross product

a=i+2j+3k, b=2i-j+k.

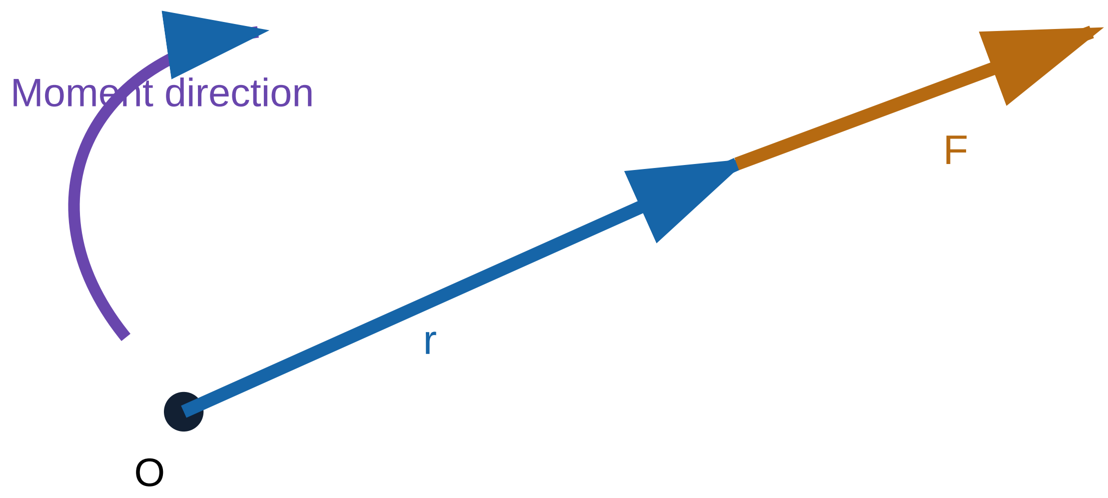
$$a\times b = 5i+5j-5k$$

|a×b|=5√3, which equals the area of the parallelogram formed by a and b.

8. Moment of a force

$$\text{Moment about O: } M = r\times F$$

Here r is the position vector from the moment centre to any point on the line of action of F.



The moment vector is normal to the plane containing r and F. Its magnitude is force × perpendicular distance.

Solved Example 13: Moment

r=2i+j m and F=3i+4j N.

$$M=r\times F=(2\times 4-1\times 3)k=5k\ N\cdot m$$

Answer: 5 N·m in the positive k direction.

Series Test I preparation
Practise one 3×3 determinant, one matrix product, one Cramer problem, one inverse problem, one dot-product/work problem and one cross-product/moment problem.

Module 2 Quick Check and Answers

1. Is speed scalar or vector? **Scalar.**
2. When are two nonzero vectors perpendicular? **When their dot product is zero.**
3. What does $|a \times b|$ represent geometrically? **Area of the parallelogram.**
4. State the SI unit of moment. **N·m.**

MODULE 3 • CO3 • 18 HOURS

Integral Calculus

Integration as inverse differentiation, standard results, substitution, integration by parts and definite integrals.

1. Integration as reverse differentiation

If $dF/dx=f(x)$, then F is an antiderivative of f and

$$\int f(x)dx = F(x)+C$$

The constant C is essential because the derivative of every constant is zero.

Malayalam Support: Differentiation-ന്റെ reverse operation ആണ് integration. Indefinite integral എഴുതുമ്പോൾ +C ഒഴിവാക്കരുത്.

Verification
Differentiate your final answer. If its derivative equals the original integrand, the integration is correct.

2. Standard integrals

Integrand	Integral	Condition / note
x^n	$x^{n+1}/(n+1)+C$	$n \neq -1$
$1/x$	$\ln\|x\|+C$	$x \neq 0$
e^x	e^x+C	—
a^x	$a^x/\ln a + C$	$a>0, a \neq 1$
$\sin x$	$-\cos x+C$	—
$\cos x$	$\sin x+C$	—
$\sec^2 x$	$\tan x+C$	—
$\mathrm{cosec}^2 x$	$-\cot x+C$	—
$\sec x \tan x$	$\sec x+C$	—
$\mathrm{cosec}\, x \cot x$	$-\mathrm{cosec}\, x+C$	—
$1/(1+x^2)$	$\tan^{-1}x+C$	—
$1/\sqrt{1-x^2}$	$\sin^{-1}x+C$	$\|x\|<1$

3. Basic rules

$$\int[af(x)\pm bg(x)]dx = a\int f(x)dx \pm b\int g(x)dx$$

Solved Example 14: Standard results

Evaluate $\int(3x^2-4x+5)dx$.

$$=x^3-2x^2+5x+C$$

Check: $d/dx(x^3-2x^2+5x+C)=3x^2-4x+5$.

4. Integration by substitution

Substitution is useful when part of the integrand is the derivative of another part. Replace the inner expression by a new variable.

1. Choose $u=g(x)$.
2. Find $du=g'(x)dx$.
3. Rewrite the entire integral in u.
4. Integrate with respect to u.
5. Substitute back and add C for indefinite integrals.

Solved Example 15: Substitution

Evaluate $\int 2x(x^2+1)^4dx$.

Let $u=x^2+1$, $du=2x\,dx$.

$$\int u^4du=u^5/5+C=(x^2+1)^5/5+C$$

Solved Example 16: Logarithmic pattern

Evaluate $\int(3x^2)/(x^3+5)\,dx$.

Let $u=x^3+5$, $du=3x^2dx$.

$$\int du/u=\ln|u|+C=\ln|x^3+5|+C$$

Common mistake

Do not substitute only the denominator. Every factor and dx must be rewritten consistently.

5. Integration by parts

$$\int u\,dv = uv - \int v\,du$$

Choose u using the practical priority ILATE: Inverse trigonometric, Logarithmic, Algebraic, Trigonometric, Exponential.

Solved Example 17: $\int x e^x dx$

Choose $u=x$, $dv=e^x dx$. Then $du=dx$ and $v=e^x$.

$$\int xe^x dx = xe^x - \int e^x dx = e^x(x-1)+C$$

Solved Example 18: $\int x \sin x\, dx$

$u=x$, $dv=\sin x\, dx$; $du=dx$, $v=-\cos x$.

$$\int x \sin x\, dx = -x \cos x + \int \cos x\, dx = -x \cos x + \sin x + C$$

6. Definite integrals

A definite integral has fixed lower and upper limits and gives a numerical value. If $F'(x)=f(x)$,

$$\int_a^b f(x)dx = [F(x)]_a^b = F(b)-F(a)$$

Malayalam Support: Definite integral-ൽ +C വേണ്ട. Upper limit substitute ചെയ്ത value-ൽ നിന്ന് lower limit value കുറയ്ക്കണം.

Limit reversal

$$\int_a^b f\, dx = -\int_b^a f\, dx$$

Same limits

$$\int_a^a f\, dx = 0$$

Interval splitting

$$\int_a^c f\, dx = \int_a^b f\, dx + \int_b^c f\, dx$$

Linearity

Constants, sums and differences can be separated as in indefinite integration.

Solved Example 19: Definite polynomial integral

Evaluate $\int_0^2 (3x^2+2x)dx$.

$$[x^3+x^2]_0^2=(8+4)-0=12$$

Answer: 12.

Solved Example 20: Definite integral by substitution

Evaluate $\int_0^1 2x(x^2+1)\,dx$.

$u=x^2+1$. When $x=0$, $u=1$; when $x=1$, $u=2$.

$$\int_1^2 u\,du=[u^2/2]_1^2=(4-1)/2=3/2$$

7. Choosing a method

Pattern	Preferred method	Example
Direct standard form	Standard result	$\int x^5 dx$
$g'(x)[g(x)]^n$ or $g'(x)/g(x)$	Substitution	$\int 2x(x^2+1)^3 dx$
Product of unlike functions	By parts	$\int xe^x dx$
Fixed limits	Find antiderivative, then apply limits	$\int_0^2 x^2 dx$

Module 3 exam tip

Before integrating, simplify algebraically and identify the pattern. A correct method choice saves most of the work.

Module 3 Quick Check and Answers

1. $\int 4x^3 dx$ = $\mathbf{x^4+C}$.
2. $\int \cos x\, dx$ = **sin x+C**.
3. Why is +C omitted in a definite integral? **It cancels in F(b)-F(a).**
4. State integration by parts. **∫u dv=uv-∫v du.**

MODULE 4 • CO4 • 12 HOURS • SERIES TEST II

Applications of Integration and Differential Equations

Area bounded by a curve and the axes; first-order, first-degree differential equations by variable separable and linear methods.

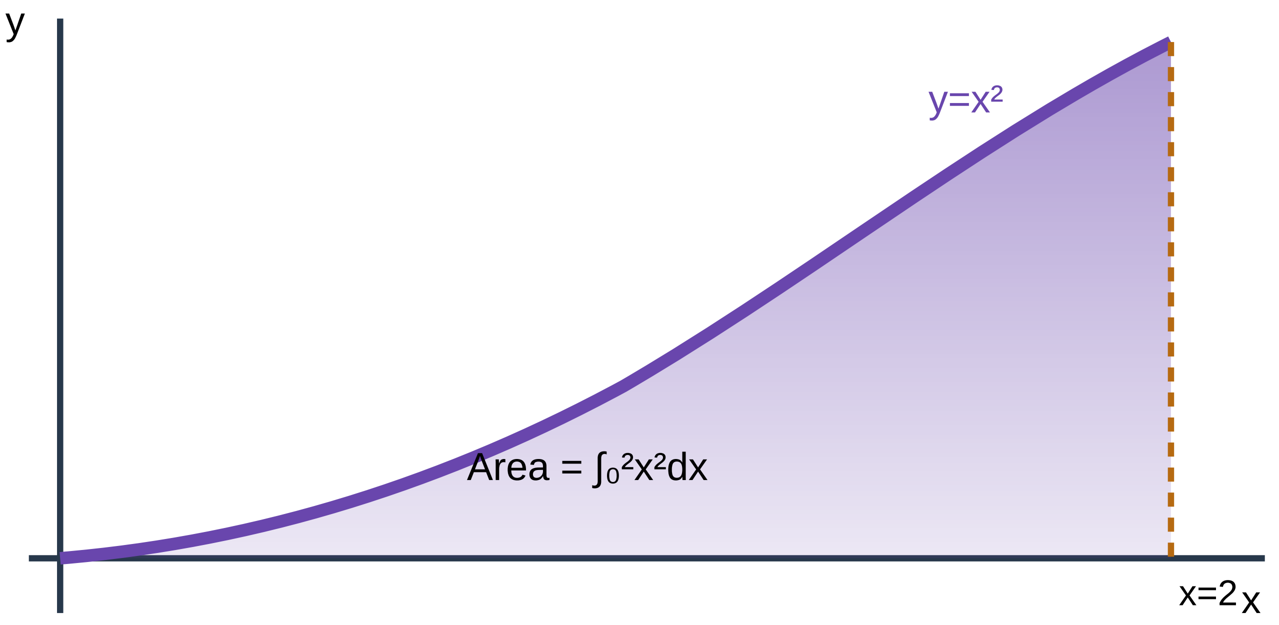
1. Area bounded by a curve and axes

If $y=f(x)$ lies above the x-axis from $x=a$ to $x=b$, the area is

$$A = \int_a^b y\,dx = \int_a^b f(x)dx$$

If the region is described more easily by $x=g(y)$, use $A=\int x\,dy$. Area is positive, so split the interval if the curve crosses an axis.

Malayalam Support: Curve-നും axis-നും ഇടയിലെ area കണ്ടെത്താൻ graph നോക്കി limits ശരിയായി എഴുതണം. Curve axis-ന്റെ താഴെയാണെങ്കിൽ integral negative ആകാം; geometric area positive ആയി എടുക്കണം.



For $y=x^2$, $x=0$, $x=2$ and the x-axis, the required area is the shaded region.

Solved Example 21: Area under a parabola

Find the area bounded by $y=x^2$, $x=0$, $x=2$ and the x-axis.

$$A=\int_0^2 x^2dx=[x^3/3]_0^2=8/3 \text{ square units}$$

Solved Example 22: Area bounded by a line and axes

The line $y=6-2x$ meets the y-axis at (0,6) and x-axis at (3,0).

$$A=\int_0^3(6-2x)dx=[6x-x^2]_0^3=18-9=9 \text{ square units}$$

Check by triangle area: $\frac{1}{2}\times3\times6=9$.

2. Differential equations: basic concepts

A differential equation contains derivatives of an unknown function. The **order** is the highest derivative order. The **degree** is the power of the highest-order derivative after the equation is polynomial in derivatives.

First order

Highest derivative is dy/dx.

First degree

The highest derivative occurs to power one.

General solution

Contains an arbitrary constant.

Particular solution

Obtained after applying an initial/boundary condition.

Solved Example 23: Order and degree

$(dy/dx)^2 + y = x$ has order 1 and degree 2. The equation $d^2y/dx^2 + 3dy/dx + y=0$ has order 2 and degree 1.

3. Variable separable equations

An equation is separable when it can be rearranged so all y terms occur with dy and all x terms occur with dx.

$$dy/dx = f(x)g(y) \Rightarrow dy/g(y) = f(x)dx$$

1. Separate variables.
2. Integrate both sides.
3. Add one arbitrary constant.
4. Simplify the relation.
5. Use initial data if provided.

Solved Example 24: Separable equation

Solve $dy/dx=xy$.

$$dy/y = x\,dx$$

$$\ln|y|=x^2/2+C \Rightarrow y=Ce^{x^2/2}$$

Solved Example 25: Initial condition

Solve $dy/dx=2xy$, $y=3$ when $x=0$.

$dy/y=2x\,dx \Rightarrow \ln|y|=x^2+C \Rightarrow y=Ce^{x^2}$.

At $x=0$, $y=3$, hence $C=3$.

Answer: $y=3e^{x^2}$.

4. Linear differential equations

The standard first-order linear form is

$$dy/dx + P(x)y = Q(x)$$

$$\text{Integrating factor (I.F.)} = e^{\int P(x)dx}$$

Solution: $y \cdot I.F. = \int Q(x) \cdot I.F.\, dx + C$

Malayalam Support: Linear equation ആദ്യം dy/dx + Py = Q എന്ന standard form-ൽ എഴുതുക. ശേഷം I.F.=e^(∫Pdx) കണ്ടെത്തി formula ഉപയോഗിക്കുക.

Solved Example 26: Linear equation

Solve $dy/dx+y=e^x$.

$P=1$, $Q=e^x$. $I.F.=e^{\int 1dx}=e^x$.

$$ye^x=\int e^x \cdot e^x dx+C=\int e^{2x}dx+C=e^{2x}/2+C$$

$$y=e^x/2+Ce^{-x}$$

Solved Example 27: Linear equation with variable coefficient

Solve $dy/dx+(1/x)y=x^2$, $x>0$.

$I.F.=e^{\int(1/x)dx}=e^{\ln x}=x$.

$$xy=\int x \cdot x^2 dx+C=x^4/4+C$$

$$y=x^3/4+C/x$$

5. Choosing the differential-equation method

Form	Method	First action
dy/dx=f(x)g(y)	Variable separable	Move y factors with dy and x factors with dx.
dy/dx+P(x)y=Q(x)	Linear	Identify P and Q; calculate I.F.
Can be rearranged into either form	Choose shorter route	Simplify before deciding.

Series Test II preparation

Practise standard integrals, one substitution, one by-parts problem, one definite integral, one area problem, one separable equation and one linear equation.

Module 4 Quick Check and Answers

1. What determines the order of a differential equation? **The highest derivative.**
2. State the I.F. for dy/dx+Py=Q. $\mathbf{e^{\int Pdx}}$.
3. Find area under y=x from 0 to 2. **2 square units.**
4. Is dy/dx=xy separable? **Yes.**

FORMULA BANK

Essential formulas with conditions

2×2 determinant

$|a\ b;\ c\ d| = ad - bc$

2×2 inverse

$A^{-1} = 1/(ad-bc)[d\ -b;\ -c\ a]$

Only if $ad-bc \neq 0$.

Cramer's rule

$x = \Delta x/\Delta,\ y = \Delta y/\Delta,\ z = \Delta z/\Delta$

Requires $\Delta \neq 0$.

Magnitude

$|a| = \sqrt{(a_x^2 + a_y^2 + a_z^2)}$

Dot product

$a \cdot b = |a||b|\cos\theta = \Sigma a_i b_i$

Cross product

$a \times b = |a||b|\sin\theta\ \hat{n}$

Work

$W = F \cdot s$

SI unit: joule.

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Moment

$$M=r\times F$$

SI unit: N·m.

Power rule

$$\int x^n dx=x^{n+1}/(n+1)+C$$

$n\neq-1$.

Log rule

$$\int dx/x=\ln|x|+C$$

By parts

$$\int u\,dv=uv-\int v\,du$$

Definite integral

$$\int_a^b f(x)dx=F(b)-F(a)$$

Area

$$A=\int_a^b y\,dx$$

Split intervals if the curve crosses an axis.

Separable DE

$$dy/g(y)=f(x)dx$$

Linear DE

$$dy/dx+Py=Q; IF=e^{\wedge}(\int Pdx)$$

METHOD BANK

Problem-solving procedures

3×3 determinant

1. Choose a row or column.
2. Write + - + signs.
3. Form each 2×2 minor.
4. Simplify carefully.

Matrix inverse solution

1. Write $AX=B$.
2. Find $|A|$.
3. Find A^{-1}.
4. Compute $X=A^{-1}B$.
5. Verify.

Cramer’s rule

1. Write Δ.
2. Form Δx, Δy, Δz.
3. Evaluate.
4. Divide by Δ.
5. Check original equations.

Vector application

1. Write components.
2. Choose dot or cross product.
3. Calculate.
4. State direction/unit.

Integration

1. Simplify.
2. Identify standard/substitution/parts.
3. Integrate.
4. Add C or apply limits.
5. Differentiate to check.

Differential equation

1. Identify separable or linear form.
2. Apply method.
3. Integrate.
4. Add C.
5. Use condition and verify.

IMPORTANT DIAGRAM AND GRAPH BANK

Visual ideas to reproduce in answers

Visual	Must show	Why it matters
Matrix multiplication	Row of first matrix, column of second, product entry	Prevents order mistakes
Parallelogram law	a, b and resultant a+b	Shows vector addition geometrically
Dot product angle	Two vectors and included angle θ	Connects product with projection/work
Cross product / moment	r, F and perpendicular moment direction	Shows right-hand rule and torque
Area under curve	Curve, axes, limits and shaded region	Helps choose correct integral
Linear DE flow	Standard form → IF → solution formula	Organises steps

SOLVED PROBLEM BANK

Additional examination-style worked examples

S1. Solve two equations by inverse method

2x+3y=13, x+2y=8.

A=[2 3;1 2], |A|=1, A^{-1}=[2 -3;-1 2].

$$X=A^{-1}B=[2\ -3;-1\ 2][13;8]=[2;3]$$

Answer: x=2, y=3.

S2. Cramer's rule

Solve $x+y+z=6$, $x-y+z=2$, $2x+y-z=1$.

$\Delta=6$, $\Delta x=6$, $\Delta y=12$, $\Delta z=18$.

Answer: $x=1$, $y=2$, $z=3$.

S3. Work by a force

$F=5i+2j-k$ N, $s=3i-j+4k$ m.

$$W=F\cdot s=15-2-4=9\text{ J}$$

S4. Moment

$r=i+2j+3k$ m, $F=2i-j+k$ N.

$$M=r\times F=5i+5j-5k\text{ N}\cdot\text{m}$$

S5. Substitution

$\int x/\sqrt{(x^2+4)}\,dx$. Let $u=x^2+4$, $du=2x\,dx$.

$$=\tfrac{1}{2}\int u^{-1/2}du=\sqrt{u}+C=\sqrt{(x^2+4)}+C$$

S6. By parts

$\int x \ln x\,dx$. Let $u=\ln x$, $dv=x\,dx$.

$$=x^2\ln x/2 - \tfrac{1}{2}\int x\,dx = x^2\ln x/2 - x^2/4 + C$$

S7. Definite integral

$\int_0^{(\pi/2)} \sin x\,dx=[-\cos x]_0^{(\pi/2)}=1$.

S8. Area

Area under $y=4-x^2$ from $x=0$ to $x=2$:

$$A=\int_0^2(4-x^2)dx=[4x-x^3/3]_0^2=16/3\text{ square units}$$

S9. Separable DE

$dy/dx=x/y$.

$$y\,dy=x\,dx \Rightarrow y^2/2=x^2/2+C \Rightarrow y^2=x^2+C$$

S10. Linear DE

$dy/dx+2y=4$.

$IF=e^{2x}$.

$$ye^{2x}=\int 4e^{2x}dx+C=2e^{2x}+C \Rightarrow y=2+Ce^{-2x}$$

EXPECTED QUESTIONS

Module-wise high-priority practice

These are fresh syllabus-based practice patterns, not a claim about the exact examination paper.

Module 1

- Define determinant, matrix and order.
- Evaluate second- and third-order determinants.
- List and identify matrix types.
- Add, subtract and multiply matrices.
- Solve three equations by Cramer's rule.
- Solve two equations using inverse matrix.

Module 2

- Differentiate scalar and vector quantities.
- Find magnitude and unit vector.
- Find vector between two points.
- Use dot product to find angle or work.
- Use cross product to find area or moment.
- Interpret product sign and direction.

Module 3

- Explain integration as reverse differentiation.
- State standard integral results.
- Solve polynomial and trigonometric integrals.
- Apply substitution.
- Apply integration by parts.
- Evaluate definite integrals.

Module 4

- Set limits from a graph.
- Find area bounded by curve and axes.
- Define order and degree.
- Solve separable equations.

- Solve linear first-order equations.
- Apply an initial condition.

PRACTICE SECTION

Objective, short-answer and problem practice

A. Objective questions

1. The determinant of [2 3;1 4] is: (a) 5 (b) 8 (c) 11 (d) -5.
2. A matrix with all diagonal entries 1 and all others 0 is: (a) zero (b) identity (c) row (d) rectangular.
3. If a·b=0 for nonzero vectors, they are: (a) parallel (b) equal (c) perpendicular (d) opposite.
4. |a×b| gives the area of a: (a) triangle (b) circle (c) parallelogram (d) rectangle only.
5. ∫dx/x equals: (a) $1/x^2$ (b) ln|x|+C (c) $x^2/2$ (d) e^x.
6. The I.F. of dy/dx+3y=x is: (a) e^x (b) e^{3x} (c) $3e^x$ (d) x^3.

B. Short problems

7. Evaluate |1 2 0; 3 -1 2; 2 1 1|.
8. Find AB for A=[1 2;0 3], B=[2 1;4 -1].
9. Find a unit vector along 2i-2j+k.
10. Find work when F=2i+3j N and s=4i-j m.
11. Evaluate $\int(4x^3+2/x)dx$.
12. Evaluate $\int_0^1(x^2+1)dx$.
13. Find area under y=2x from x=0 to x=3.
14. Solve $dy/dx=3x^2y$.
15. Solve dy/dx+y=1.

C. Long/application problems

16. Solve x+y+z=6, 2x+y-z=1, x-y+2z=5 by Cramer’s rule.
17. Solve 4x+y=11, 2x+3y=13 using inverse matrix.
18. For r=2i-j+k and F=i+3j-2k, find moment and magnitude.
19. Evaluate $\int x^2e^xdx$ using repeated integration by parts.
20. Find the area bounded by y=x(4-x), the x-axis, x=0 and x=4.

Practice final answers

1. 5
2. Identity
3. Perpendicular
4. Parallelogram
5. ln|x|+C
6. e^{3x}
7. -1
8. [10 -1;12 -3]
9. (2i-2j+k)/3
10. 5 J
11. $x^4+2\ln|x|+C$
12. 4/3
13. 9 square units
14. $y=Ce^{x^3}$
15. $y=1+Ce^{-x}$
16. x=1, y=2, z=3
17. x=2, y=3
18. M=-i+5j+7k; $|M|=\sqrt{75}$
19. $e^x(x^2-2x+2)+C$
20. 32/3 square units

Self-assessment

- ✓ I can identify the order and type of a matrix.
- ✓ I can evaluate a 3×3 determinant without sign errors.
- ✓ I can decide between dot and cross product.
- ✓ I can choose standard, substitution or by-parts integration.
- ✓ I can read integration limits from a graph.
- ✓ I can distinguish separable and linear differential equations.

QUICK REVISION

Last-day revision sheet

Determinant
2×2: ad-bc; 3×3: + - + expansion.

Inverse
Exists only if determinant is nonzero.

Matrix product
Inner dimensions must match.

Dot product
Scalar result; angle and work.

Cross product
Vector result; area and moment.

Integration
Reverse differentiation; include +C.

Definite integral
F(b)-F(a), no +C in final value.

Area
Sketch, select limits, integrate positive region.

Separable DE
Separate x and y before integrating.

Linear DE
Standard form, I.F., then solution formula.

Ten common mistakes

Wrong cofactor sign; changing variable order in Cramer's rule; multiplying matrices entry-by-entry; forgetting $|A|\neq 0$; confusing dot and cross products; omitting units for work/moment; forgetting +C; incorrect substitution differential; applying limits in reverse; using the linear-DE formula before standardising the equation.

FRESH SYLLABUS-BASED MODEL

Model Question Paper

The paper follows the common 75-mark diploma theory structure and is newly written from the official Mathematics II syllabus.

MATHEMATICS II - 2002

Time: 3 Hours | Maximum Marks: 75

Instructions: Answer as directed. Show essential working. Assume suitable missing data only where necessary. Use exact values unless approximation is requested.

PART A — 5×1=5

Answer all questions.

1. Evaluate |3 1;2 4|.
2. State the condition for the inverse of a square matrix to exist.
3. What is the dot product of perpendicular nonzero vectors?
4. Write $\int \cos x\, dx$.
5. State the integrating factor of $dy/dx+P(x)y=Q(x)$.

PART B — 5×8=40

Answer any five questions.

6. (a) Evaluate |2 1 3;1 0 2;3 2 1|. (b) Add and multiply A=[1 2;3 1], B=[2 0;1 4].
7. Solve x+y+z=6, 2x-y+z=3, x+2y-z=2 by Cramer's rule.
8. For a=2i-j+2k and b=i+3j-k, find a·b, a×b and the angle between them.
9. A force F=3i+4j-2k N acts at position r=2i-j+k m. Find the moment about the origin and its magnitude.
10. Evaluate (a) $\int 2x(x^2+4)^3dx$ and (b) $\int xe^x dx$.
11. Evaluate (a) $\int_0^2(x^2+2x)dx$ and (b) find the area under y=3x from x=0 to x=2.
12. Solve (a) dy/dx=xy and (b) dy/dx+2y=6.

PART C — 2×15=30

Answer any two questions.

13. (a) Explain important matrix types with examples. (b) Solve 3x+y=7 and 2x+y=5 by inverse method. (c) Verify the answer.
14. (a) Explain scalar and vector products. (b) A force F=4i-2j+3k N moves a point through s=2i+j-4k m; find work. (c) At r=i+2j m, a force 3i+4j N acts; find moment.
15. (a) Evaluate $\int x \sin x\, dx$. (b) Find the area bounded by $y=4-x^2$, the x-axis and x=0 in the first quadrant. (c) Solve $dy/dx+(1/x)y=x^2$, x>0.

COMPLETE MODEL ANSWERS

Full procedures and final answers

Part A

1. $3\times4-1\times2=10$.

2. The determinant must be nonzero: $|A|\neq0$.

3. Zero.

4. $\sin x+C$.

5. $e^{\int P(x)dx}$.

Part B

6. Determinant and matrix operations

(a) $\Delta=2(0\times1-2\times2)-1(1\times1-2\times3)+3(1\times2-0\times3)=-8+5+6=3$.

(b) $A+B=[3\ 2;4\ 5]$.

$$AB=[1\times2+2\times1,\ 1\times0+2\times4;\ 3\times2+1\times1,\ 3\times0+1\times4]=[4\ 8;7\ 4]$$

7. Cramer’s rule

$\Delta=7$, $\Delta x=7$, $\Delta y=14$, $\Delta z=21$.

Answer: $x=1$, $y=2$, $z=3$. Substitution satisfies all three equations.

8. Vector products and angle

$a\cdot b=2\times1+(-1)\times3+2\times(-1)=-3$.

$$a\times b = -5i+4j+7k$$

$|a|=3$, $|b|=\sqrt{11}$. $\cos\theta=-3/(3\sqrt{11})=-1/\sqrt{11}$.

Answer: $\theta=\cos^{-1}(-1/\sqrt{11})$.

9. Moment

$$M=r\times F=\begin{bmatrix} i & j & k \\ 2 & -1 & 1 \\ 3 & 4 & -2 \end{bmatrix}=-2i+7j+11k$$

$|M|=\sqrt{(4+49+121)}=\sqrt{174}$ N·m.

10. Integration

(a) $u=x^2+4$, $du=2x\,dx$.

$$\int u^3 du = u^4/4 + C = (x^2+4)^4/4 + C$$

(b) $u=x$, $dv=e^x dx$.

$$\int x e^x dx = xe^x - e^x + C = e^x(x-1) + C$$

11. Definite integral and area

(a) $[x^3/3+x^2]_0^2=8/3+4=20/3$.

(b) $A=\int_0^2 3x\,dx=[3x^2/2]_0^2=6$ square units.

12. Differential equations

(a) $dy/y=x\,dx \Rightarrow \ln|y|=x^2/2+C \Rightarrow y=Ce^{x^2/2}$.

(b) $IF=e^{2x}$. $ye^{2x}=\int 6e^{2x}dx+C=3e^{2x}+C$. Hence $y=3+Ce^{-2x}$.

Part C

13. Matrices and inverse method

(a) A row matrix has one row; column matrix one column; rectangular matrix unequal rows and columns; square matrix equal rows and columns; zero matrix all entries zero; diagonal matrix non-diagonal entries zero; scalar matrix equal diagonal entries; identity matrix diagonal entries one and other entries zero.

(b) $A=[3\ 1;2\ 1]$, $X=[x;y]$, $B=[7;5]$. $|A|=1$.

$$A^{-1}=[1\ -1;-2\ 3],\ X=A^{-1}B=[2;1]$$

(c) Verification: $3(2)+1=7$ and $2(2)+1=5$. Therefore $x=2, y=1$.

14. Scalar/vector products, work and moment

(a) Dot product $a\cdot b=|a||b|\cos\theta$ is a scalar and is used for angle, projection and work. Cross product $a\times b=|a||b|\sin\theta\ \hat{n}$ is a vector perpendicular to both and is used for area and moment.

(b) $W=F\cdot s=4\times2+(-2)\times1+3\times(-4)=-6$ J.

(c) $r= i+2j$, $F=3i+4j$.

$$M=r\times F=(1\times4-2\times3)k=-2k\ \text{N}\cdot\text{m}$$

The negative k direction indicates clockwise sense in the xy-plane.

15. Integration, area and linear differential equation

(a) $u=x, dv=\sin x\,dx$. Then $v=-\cos x$.

$$\int x \sin x\,dx = -x\cos x + \sin x + C$$

(b) In the first quadrant, $y=4-x^2$ meets x-axis at $x=2$.

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$$A=\int_0^2(4-x^2)dx=[4x-x^3/3]_0^2=16/3 \text{ square units}$$

(c) $P=1/x$, $IF=e^{\int dx/x}=x$.

$$xy=\int x\cdot x^2dx+C=x^4/4+C \Rightarrow y=x^3/4+C/x$$

ANSWER KEY

Compact answer index

Area	Key answers / checks
Module 1	Determinant sign pattern + - +; AB exists when inner dimensions match; inverse exists for $\|A\|\neq0$.
Module 2	$a\cdot b$ scalar; $a\times b$ vector; work=$F\cdot s$; moment=$r\times F$.
Module 3	Always add C to indefinite integrals; definite integral uses upper minus lower.
Module 4	Area needs graph and limits; separable equations separate variables; linear equations use IF.
Practice MCQ	1-a, 2-b, 3-c, 4-c, 5-b, 6-b.

OFFICIAL REFERENCES

Textbooks and online resource guidance

Code	Reference
T1	B. S. Grewal, *Higher Engineering Mathematics*, Khanna Publishers, New Delhi, 40th Edition, 2007.
R1	G. B. Thomas and R. L. Finney, *Calculus and Analytic Geometry*, Addison Wesley, 9th Edition, 1995.
R2	Reena Garg, *Engineering Mathematics*, Khanna Publishing House, New Delhi, Revised Edition, 2018.
R3	S. S. Sabharwal and Sunita Jain, *Applied Mathematics, Vol. I & II*, Eagle Parkashan, Jalandhar.

Online resources listed by the syllabus

Use e-books, e-tools, learning mathematics software and suitable mathematics websites for additional visualisation and practice. The official syllabus does not specify individual web addresses.

FINAL EXAM REVISION CHECKLIST

Before entering the examination hall

Concept checklist

- ✓ Determinant order and 3×3 sign pattern
- ✓ Matrix types and multiplication condition
- ✓ Dot versus cross product
- ✓ Standard integrals and method selection
- ✓ Area limits from graph
- ✓ Separable versus linear DE

Presentation checklist

- ✓ Write formula before substitution
- ✓ Keep variable order consistent
- ✓ Show vector units and direction
- ✓ Add +C where required
- ✓ Apply upper minus lower limit
- ✓ Box the final answer and verify

Final reminder

Mathematics marks are earned through correct method and visible working. Even when the final arithmetic is wrong, a clearly stated formula and correct intermediate steps may receive partial credit.

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Standalone student handbook prepared from the official syllabus. Use the Download as PDF button for an offline A4 copy.